



OTC Desk Agent Arena — Run #134

Gemini 3.8 Flash is the more accurate model and finishes behind the one it replaces. The extra spend is not retries and not subagent fan-out: it is five times as much filesystem searching, at the same hit rate, in the same places. A control run says roughly four fifths of that is the model and one fifth is a change we made to the harness the week before.

2026-09-03 · run #134 · 1 model × 6 workflows × 2 trials (5 workflows) and 1 trial (confirmation-desk-day) · pinned to xhigh · output budget unpinned · objective-only scoring (jury off) · 6 of 6 arms scored, zero truncated calls, zero malformed tool calls · compared against the gemini-3-7-flash arm of run #127, the same route at the same effort · control arm: run #135

Why this run exists

Gemini 3.8 Flash was published on 2026-09-01. The desk's previous Gemini contestant, 3.7 Flash, holds an unusual position in the standings: it has posted the **highest objective score in the flash tier** (94.3 on run #129) while ranking third, because its efficiency is the worst of the leading group. The obvious question for a successor was whether it kept the accuracy and fixed the cost.

It did the opposite of the second half.

Headline

Consolidated card, six workflows, published to /arena/models.html as cards only — one model has no field, so a rank of #1 of 1 would measure nothing.

Model	eff	OVR	GRD	ADH	SYN	PRC	EFF	CON	Objective
Gemini 3.8 Flash	xhigh	79	91	97	99	95	4	89	95.6
Gemini 3.7 Flash (run #129)	xhigh	83	95	92	99	92	31	89	94.3

On the five workflows the two arms share, at the same effort on the same route:

	Gemini 3.8 Flash	Gemini 3.7 Flash
Objective score	97.1	94.3
Workflows won on objective	4 of 5	1 of 5
Perfect 100.0 scores	2	0
Tool calls per trial	95.9	52.0
EFF	4	31
Mean OVR	80.6	83.0

The newer model is better at the work and ranks lower for it. It wins four of the five workflows on correctness, posts two perfect scores where its predecessor posts none, and spends **84% more tool calls** to do it. EFF is scored as golf against each workflow's calibrated par, so that difference is not a rounding adjustment — it is the whole ranking.

Workflow	3.8 obj	3.8 calls	3.7 obj	3.7 calls
ops-settlement-day	100.0	84.5	89.8	38.5
risk-limit-breach-day	100.0	45.0	98.7	29.0
risk-manager-control-day	97.4	92.0	93.6	58.5
trader-rfq-booking-day	96.8	136.5	93.7	69.5
high-board-portfolio-review-day	91.4	121.5	95.7	64.5

Every row goes the same way. There is no workflow where 3.8 is both more accurate and cheaper, and none where it is cheaper at all.

Where the calls actually go

The scores say *how much* more it spends. The transcripts say *on what*. Counting every tool call in all ten trials across the five shared workflows, then averaging per trial:

Tool	3.8 Flash	3.7 Flash	Delta
<code>grep</code>	64.5	14.5	+50.0
<code>ls</code>	22.5	0	+22.5
<code>glob</code>	23.0	6.5	+16.5
<code>read_artifact</code>	17.5	2.5	+15.0
<code>inspect_artifact</code>	11.5	2.5	+9.0
<code>list_artifacts</code>	10.0	3.0	+7.0

Grouping those together against everything else:

	3.8 Flash	3.7 Flash
Filesystem / search calls	149	29
Domain and quant tool calls	352	253
Search as a share of all calls	29.7%	10.3%

Filesystem search is 55% of the entire increase. The remaining 45% is spread thinly over two dozen domain tools, none of which moves by more than about ten calls. The single most striking cell in the table is `ls` : across ten trials and five workflows, Gemini 3.7 Flash never calls it once, and 3.8 calls it 22 times.

Where it searches is consistent. Ranked by target, 3.8's search calls go to `/skills` (53), the filesystem root (25), `/large_tool_results` (24) and `/artifacts` (8). Its predecessor touches `/skills` 24 times and almost nothing else.

What it is not

Three cheaper explanations are available, and the transcripts rule out all three.

It is not retrying after failures. Gemini 3.8 Flash records *fewer* errors than its predecessor, 0.3 per workflow against 0.8. Whatever the extra calls are, they are not recovery.

It is not subagent fan-out. Both models dispatch essentially the same number of `task()` subagents, 1.5 against 1.4 per workflow. The extra work is happening in the orchestrator's own loop.

It is not worse searching. This is the one that matters, and it is the one that survives contact with the data least comfortably for the intuitive story. Classifying every search call by what came back:

	3.8 Flash	3.7 Flash
Returned content	78%	79%
Returned nothing	17%	14%
Refused (foreign store)	5%	3%
Exact repeat of an earlier call	23%	21%

The hit rates are the same. The repeat rates are the same. Both models also concentrate their searching **late** in an episode rather than front-loading it as orientation — 55% of 3.8's search calls fall in the last third, against 60% for 3.7.

So this is not a model that searches badly, or searches differently, or searches in the wrong places. It searches *exactly the way its predecessor does*, with the same productivity per call, and does five times as much of it before it stops. The deficit is not in the searching. It is in the stopping.

The confound, and the control that sizes it

There is a problem with everything above, and it is ours rather than the model's.

The 3.7 baseline ran on 2026-08-26. Run #134 ran on 2026-09-02, and in between we changed the harness. Gemini 3 routes return an encrypted **thought signature** beside every tool call and require it echoed back on the assistant turn carrying that call; ChatOpenAI does not preserve that field, by design, so the second model turn of every tool loop was being refused with a 400. Run #134 initially failed on all six workflows before we fixed it. (The control that established this was a harness defect rather than a new-model limitation: 3.7 Flash, which has four published boards on this route, fails identically once the field is stripped.)

That fix hands the model back its own prior reasoning. It is entirely plausible that doing so makes a model more willing to keep pulling on a thread — which would mean we had measured our own change and attributed it to Google.

So we re-ran **Gemini 3.7 Flash today, with the fix**, on `ops-settlement-day`. That workflow is the right instrument: 3.7's own trial-to-trial spread there is a single call, so one trial separates the hypotheses cleanly.

Counting the transcripts, so that total calls and search calls come from one instrument:

Arm	Tool calls	Search calls	Objective
3.7 Flash, run #127, without the fix	45, 46	1, 4	89.8
3.7 Flash, run #135, with the fix	55	10	90.9
3.8 Flash, run #134, with the fix	85, 98	29, 43	100.0

The fix is not inert. It moves 3.7 from 45.5 calls to 55, and its search calls from 2.5 to 10. Had we published the naive comparison, part of what we called a model regression would have been our own patch.

It does not, however, come close to explaining the gap:

Component	Calls	Share of the gap
Total gap, 3.8 with fix vs 3.7 without	+46.0	100%
Attributable to the harness change	+9.5	21%
Attributable to the model	+36.5	79%

The same decomposition holds for search calls alone (22% harness, 78% model), and the two independent instruments — the trace harvest that feeds EFF, and a direct count of the transcripts — agree on the split to the percentage point.

The conclusion is robust to the correction. EFF is scored on the trace harvest, which puts 3.8 at 84.5 calls on this workflow. Strip the entire harness component out and it still runs `ops-settlement-day` in 75 calls against a calibrated par of **30** — $2.5\times$ par, which floors EFF just as thoroughly as $2.8\times$ does. No plausible adjustment for the harness rescues the ranking.

Caveats

- **The control is one workflow and one trial.** ZenMux subscription quota ran out mid-investigation, so the paired arm on `trader-rfq-booking-day` was deliberately abandoned to preserve what remained. The 21% figure is measured on `ops-settlement-day` and is **not** established as a constant across workflows. It is sized, not solved.
- **A comparability break now exists for Gemini contestants**, in the same way run #115 created one for Anthropic-protocol models when the output budget was raised. Every

Gemini board from #134 onward runs with the thought signature preserved; #113, #127, #129 and #130 did not, and the control says that is worth roughly a fifth of the call count on at least one workflow. Read Gemini EFF across that boundary with care.

- **confirmation-desk-day carries one trial, not two.** Its second was lost to the same quota exhaustion and recorded `invalid / infra_blank` rather than scored, so its CON is *not measured*, not perfect. That workflow is also excluded from every 3.7 comparison in this report, because 3.7 has no arm on it at pinned effort.
- **Objective scores are not adjusted for the harness change at all.** The correction above applies to call counts. Whether the restored signature also helped 3.8's accuracy is untested, and the 3.7 control moved only 1.1 points on that axis.
- Zero truncation and zero malformed tool calls, **measured** rather than assumed: `trials_measured` equals `trials_total` on all six arms.

What would change the verdict

- **A paired `low` arm.** Run #130 showed four of seven models score *higher* at `low` than at their ceiling, because lower effort cuts tool calls and EFF has the widest spread of any axis. Gemini 3.8 Flash is the most over-executing contestant the desk has measured, which makes it the likeliest beneficiary in the field. This report says nothing about how it behaves anywhere but `xhigh`.
- **The rest of the harness control.** Four more paired 3.7 arms would turn the 21% from a single-workflow measurement into a real coefficient, and would settle whether the effect is uniform or concentrated in workflows with large artifact trees.
- **A prompt-level test of the stopping rule.** The finding is that the model does not know when to stop searching, not that it searches badly. If that is right, it should be correctable from the orchestrator prompt rather than requiring a different model — and a cheap A/B on `ops-settlement-day` would show it.

Cards for this run are published at `/arena/models.html`. Nothing here reaches the leaderboard: a rank is relative and this run had no field.
